

Parameter model

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \quad a_{x_{ram}}[k] \approx a'_x[k] x_{ram}[k]$$

$$a'_x := \frac{da_x}{dh} = -\frac{a_x K_h}{R}, \quad K_h = \frac{1}{c} \frac{dc}{d\bar{h}}$$

Assumption ($x = x_{det} + x_{ram}$, $x_{det} = x_d$): $\delta x = x_d - x = -x_{ram}$, $\delta h = (x - x_d) \cdot \hat{w} = -\delta x$.

$$a_{x_{ram}} = a_x(h_d + \delta h) - a_x(h_d) = a'_x \delta h + O(\frac{1}{2} a''_x \delta h^2)$$

$$a_{x_{ram}} = a'_x x_{ram} = -a'_x \delta x$$

Gain-fluctuation variance

$$\text{Var}(a_{x_{ram}}) = a_x'^2 \text{Var}(x_{ram}) = a_x'^2 \text{Var}(\delta x) \quad (1)$$

Closed-loop ($\delta x[k+1] = \lambda_c \delta x[k] - \varepsilon[k]$, $\hat{a}_x = a_x$):

$$\text{Var}(\delta x) = C_{\delta x} \sigma_{\delta h}^2, \quad C_{\delta x} = 2 + \frac{1}{1 - \lambda_c^2}, \quad \sigma_{\delta h}^2 := 4k_B T a_{\perp}$$

Measurement ($\delta x_m = \delta x[k-d] + n_x$): $\text{Var}(\delta x_m) = \text{Var}(\delta x) + \sigma_n^2$.

$$\text{Var}(a_{x_{ram}}) = a_x'^2 (\text{Var}(\delta x_m) - \sigma_n^2) \quad (2)$$

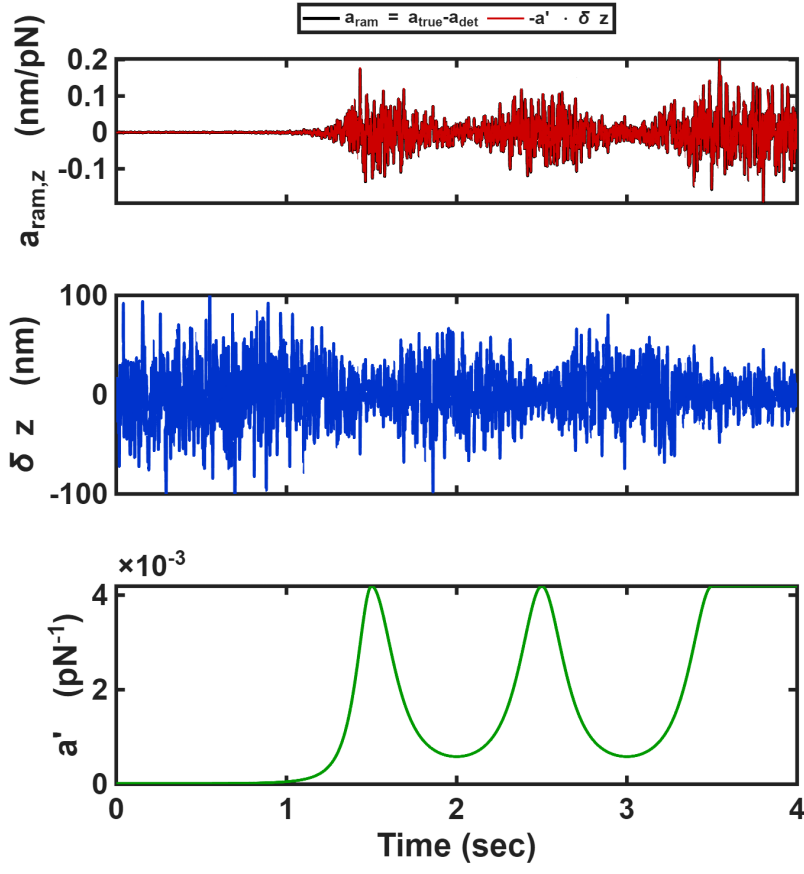
Explicit ($a_x'^2 = a_x^2 K_h^2 / R^2$):

$$\text{Var}(a_{x_{ram}}) = C_{\delta x} 4k_B T \frac{K_h^2 a_x^3}{R^2} \quad (3)$$

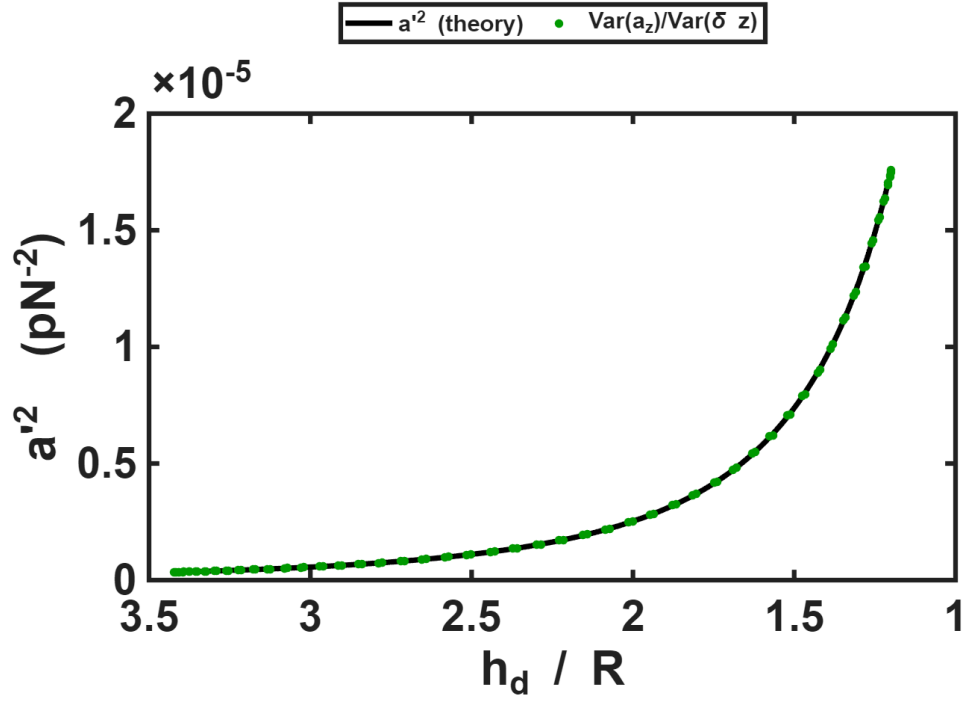
Verification

(1 Hz, near wall, cross-seed ensemble; 4-state AR(1); z ; $N = 100$)

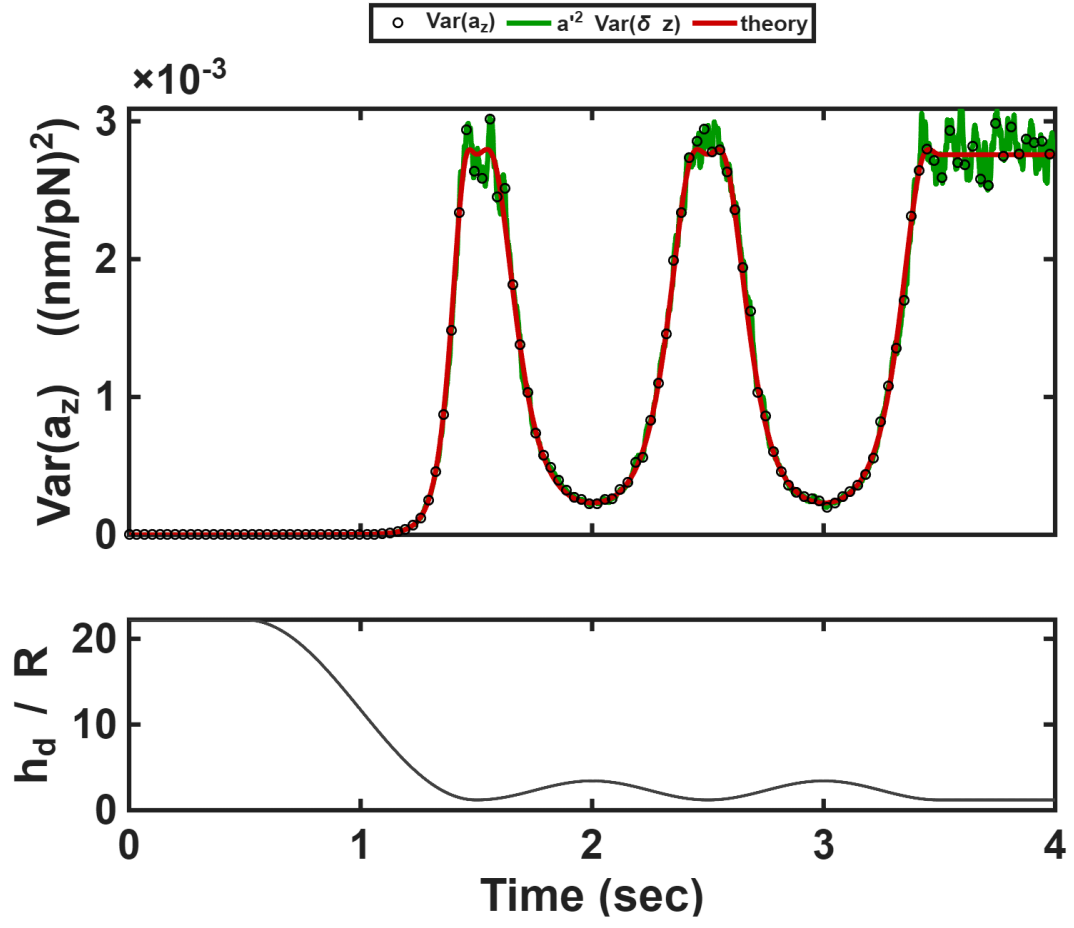
$$\delta x[k] = x_d[k] - x[k-1]$$



$a_{ram} = a_{true} - a_{det}$ and $-a' \delta z$ (top); δz (middle); a' (bottom).



$\text{Var}(a_z)/\text{Var}(\delta z)$ vs $a'^2(\bar{h})$ (form 1).



$\text{Var}(a_z)$: measured, $a'^2 \text{Var}(\delta z)$ (1), $C_{\delta x} 4k_B T K_h^2 a_z^3 / R^2$ (3); h_d / R below.